



TRIOS CYBER

LABORATORY PRACTICE 13

Correlation Rule Design and Manual Detection Testing

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1. Splunk Service Startup and Web Access

Splunk Enterprise was started on the Ubuntu laboratory virtual machine from /opt/splunk/bin. The startup checks completed successfully, the splunkd service was launched, and Splunk Web became available on TCP port 8000. This confirmed that the SIEM platform was operational before the correlation exercise was performed.

```
matthew@matthew-VirtualBox:~$ sudo /opt/splunk/bin/splunk start --run-as-root
Splunk> The IT Search Engine.

Checking prerequisites...
  Checking http port [8000]: open
  Checking mgmt port [8089]: open
  Checking appserver port [127.0.0.1:8065]: open
  Checking kvstore port [8191]: open
  Checking configuration... Done.
New certs have been generated in '/opt/splunk/etc/auth'.
  Checking critical directories... Done
  Checking indexes...
    Validated: _audit_configtracker_dm_summary_dsappevent_dsclient
    nt_dsphonehome_internal_introspection_metrics_metrics_rollup_telemetry_th
    efishbucket history main summary
    Done
  Checking filesystem compatibility... Done
  Checking conf files for problems...
    Done
  Checking default conf files for edits...
  Validating installed files against hashes from '/opt/splunk/splunk-10.4.
  3-4174a2deda5d-linux-amd64-manifest'
  All installed files intact.
```

```
Done
All preliminary checks passed.
Starting splunk server daemon (splunkd)...
Done

Waiting for web server at http://127.0.0.1:8000 to be available.....
..... Done

If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://matthew-VirtualBox:8000
matthew@matthew-VirtualBox:~$
```

2. Failed Authentication Event Review

The collected failed-login records indexed in Splunk were filtered with result=FAILED. The user and source-address fields were extracted with rex, and the events were ordered chronologically with _time. Six failed authentication events were available for manual review across the users matthew, admin, and test.

The screenshot shows the Splunk Enterprise web interface. A search is performed with the following query:

```
index=main "result=FAILED"
| rex "user=(?<user>S+)"
| rex "src=(?<src>S+)"
| table _time user src
| sort _time
```

The search results show 6 events. The table below represents the data shown in the screenshot:

_time	user	src
2026-09-12 13:00:01	matthew	10.10.10.20
2026-09-12 13:00:05	matthew	10.10.10.20
2026-09-12 13:00:10	admin	10.10.10.30
2026-09-12 13:00:20	admin	10.10.10.30
2026-09-12 13:00:25	test	10.10.10.40
2026-09-12 13:00:30	matthew	10.10.10.20

3. Candidate Event Pattern

The event list showed a repeated pattern for the same account and source address. The user matthew generated failed events from 10.10.10.20 at 13:00:01, 13:00:05, and 13:00:30. The admin account appeared twice from 10.10.10.30, while test appeared once from 10.10.10.40. These observations were used to define a threshold-based detection rule.

4. Detection Logic Definition

A simple correlation rule was defined to identify repeated failed authentication attempts. The rule triggers when three or more failed events are associated with the same username and source IP address within a maximum window of 60 seconds. This threshold provides a clear example of how multiple related events can be converted into a single detection condition.

Rule Element	Configured Value	Purpose
Event condition	result=FAILED	Restricts analysis to unsuccessful authentication events
Grouping fields	user + src	Correlates activity by account and source address
Threshold	3 or more failures	Requires repeated activity before triggering
Time window	60 seconds or less	Limits correlation to a short burst of failures
Detection result	Trigger / no trigger	Separates matching patterns from lower-frequency activity

5. Correlation Rule Test in Splunk

The logic was tested directly against the sample data with a Splunk search. The search grouped failed events by user and source IP, calculated the first and last failure timestamps, derived the event-window duration, and retained only groups with at least three failures occurring within 60 seconds.

The screenshot shows the Splunk Enterprise interface. A new search query is entered in the search bar:

```
index=main "result=FAILED"
| rex "user=(?<user>\S+)"
| rex "src=(?<src>\S+)"
| stats count as failures min(_time) as first_failure max(_time) as last_failure by user src
| eval window_seconds=last_failure-first_failure
| where failures>=3 AND window_seconds<=60
| convert ctime(first_failure) ctime(last_failure)
| table user src failures first_failure last_failure window_seconds
```

The search results are displayed in a table with the following columns: user, src, failures, first_failure, last_failure, and window_seconds. The results show one group for user 'matthew' from source IP '10.10.10.20' with 3 failures, first failure at 09/12/2026 13:00:01, last failure at 09/12/2026 13:00:30, and a window of 29 seconds.

6. Detection Result Interpretation

The rule returned one matching group: user matthew from source 10.10.10.20. Three failed authentication events occurred between 13:00:01 and 13:00:30, producing a 29-second correlation window. Because the count met the threshold and the calculated window remained below 60 seconds, the detection condition was satisfied.

User	Source IP	Failures	Observed Window	Rule Result
matthew	10.10.10.20	3	29 seconds	Triggered
admin	10.10.10.30	2	10 seconds	Not triggered
test	10.10.10.40	1	Single event	Not triggered

7. Saved Detection Report

The completed correlation search was saved in Splunk as a report named Repeated Failed Authentication Detection. The report description documented the detection condition as three or more failed authentication attempts from the same user and source IP within 60 seconds. Saving the search preserved the rule logic and the validated result for later review.

user	src	failures	first_failure	last_failure	window_seconds
matthew	10.10.10.20	3	09/12/2026 13:00:01	09/12/2026 13:00:30	29

8. Correlation Rule Summary

The manual test demonstrated the complete correlation workflow: failed events were selected, relevant fields were extracted, related records were grouped, a frequency threshold and time window were applied, and the matching activity was retained as a detection result. The rule identified only the matthew / 10.10.10.20 pattern because the other user-source combinations did not reach the configured threshold.

Field	Observed Value	Interpretation
Platform	Splunk Enterprise	Local SIEM/log-analysis platform
Analyzed events	6 failed authentication events	Input set used for rule testing
Rule threshold	>= 3 failures	Minimum count required to trigger
Correlation key	Same user + same source IP	Links related authentication events
Time window	<= 60 seconds	Maximum duration allowed for the correlated burst
Triggered pattern	matthew / 10.10.10.20	3 failures in 29 seconds
Saved report	Repeated Failed Authentication Detection	Reusable saved correlation search

9. Conclusions

The correlation-rule design exercise was successfully completed in Splunk using the collected failed-login sample events. A detection condition requiring three or more failures from the same username and source IP within 60 seconds was defined and tested manually. The sample data triggered the rule for matthew from 10.10.10.20, where three failures occurred within 29 seconds, while lower-frequency patterns did not trigger. The completed search was saved as a report, demonstrating how repeated authentication events can be converted into a simple reusable SIEM detection rule.